CircuitBuilder: From Polynomials to Circuits via Reinforcement Learning

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Introduction

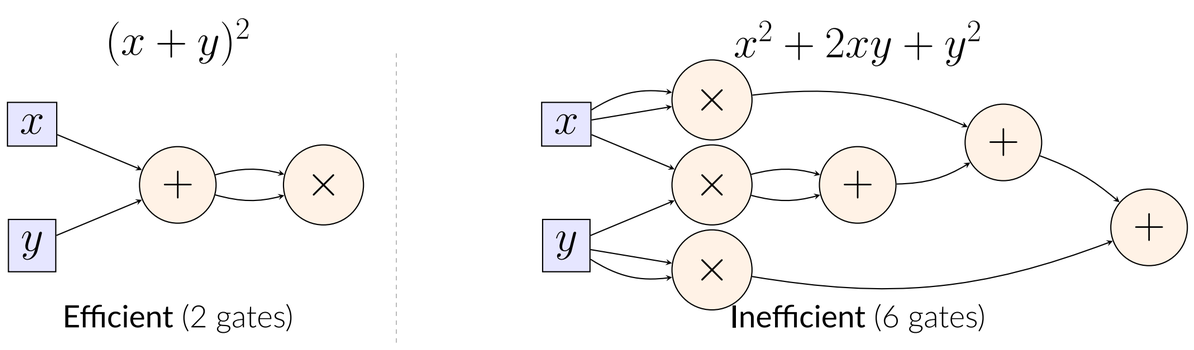
Arithmetic circuits compute polynomials using a sequence of $+$ and $\times$ gates applied to input variables. Determining the *smallest* such circuit for a given polynomial is a fundamental open problem in algebraic complexity theory, with deep connections to algorithm design and the **VP vs. VNP conjecture**—the algebraic analogue of P vs. NP. Deep learning often provides efficient solutions to NP and VNP hard problems. In this project, we train an RL agent to efficiently solve the arithmetic circuit problem.

- **Goal.** Given $f(x_1, \ldots, x_n)$, find a circuit using the minimum number of $+$ and $\times$ gates to compute it exactly.
- **Analogy.** Just as a proof is a sequence of deductions from axioms, a circuit is a sequence of operations from input variables— circuit synthesis directly mirrors proof search.
- **Challenge.** With k intermediate polynomials computed, there are $O(k^2)$ possible next operations, leading to exponential blowup in the search space.

Problem Formulation

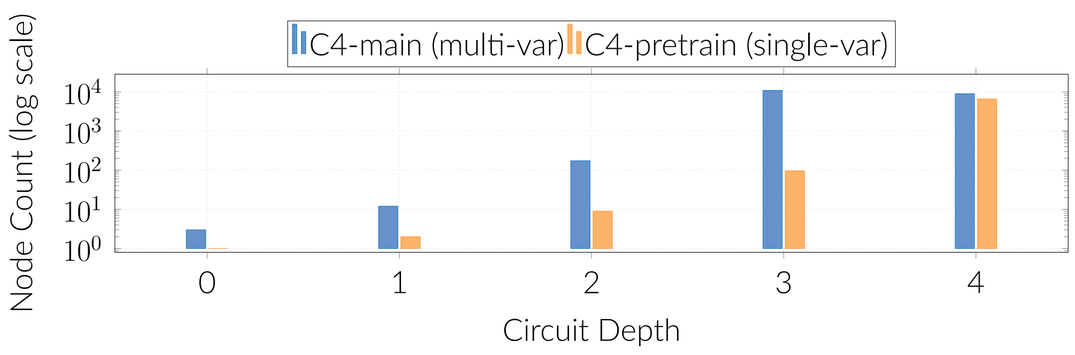
We model circuit construction as a **single-player MDP**. The problem can be viewed as a simplified version of auto-proof generation, where an agent applies a sequence of lemmas to prove a mathematical theorem.

- **State.** Computed set $\{p_1, \ldots, p_k\}$, seeded with $x_1, \ldots, x_n$ and 1.
- **Action.** Pick p_i, p_j and $\mathrm{op} \in \{+, \times\}$; append $p_{k+1} = p_i \ \mathrm{op} \ p_j$.
- **Termination.** Win when $p_k = f$; fail after N steps.



Gameboard Structure

The **C4 gameboard** is a precomputed DAG of reachable polynomials through depth 4; 94–99% of nodes admit multiple optimal circuits, confirming the vastness of the search space.



Architectural Approaches

Both agents share a **GNN-Transformer backbone**: GCNConv layers encode the circuit DAG, fused with a target polynomial embedding to produce policy logits and a value estimate.

- **SAC (off-policy).** Twin Q-heads; entropy bonus for sustained exploration; replay buffer with Polyak target updates. Complexity *decreases* if success rate falls below a lower threshold, preventing stalling.
- **CircuitBuilder (PPO+MCTS).** PPO trains the network; MCTS (64 simulations) deepens search via $\mathrm{UCB}(s,a) = Q(s,a)/N(s,a) + C\sqrt{\ln N(s)/N(s,a)}$. With p=0.35, MCTS overrides policy sampling during rollout, teaching the policy to match deeper look-ahead.

Reward Design

- **Success bonus.** $+100$ for computing the exact target.
- **Out-degree bonus.** Reward for reusing computed polynomials.

Factor Library & Reward Shaping. The *factor library* dynamically shapes rewards and introduces subgoals.

- **Divisibility.** If $p \mid g$, grant a reward and set the new subgoal to $q = g/p$.
- **Factorable residual.** If $g - p = \prod_i f_i$ factors non-trivially, grant a reward and add each f_i to the active subgoal set.

Target: $g = x^2 + 3x + 2 = (x+1)(x+2)$ • Agent computes: $p = x + 1$

↓ Two algebraic checks on p vs. g

Case 1: Divisibility
$p \mid g$ Reward ✓
New subgoal:
$g/p = x + 2$

Case 2: Residual
$g - p = (x+1)^2$ Reward ✓
New subgoals:
$\{x+1,\ x+1\}$

Key insight: Algebraic decomposition converts a sparse terminal signal into a structured sequence of tractable subgoals. This is analogous to auto-proof generation, where an agent breaks down a goal into a sequence of subgoals.

Training Pipeline

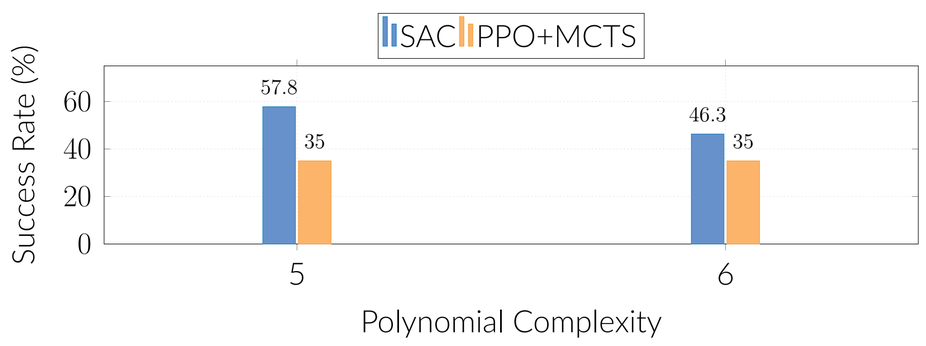
Two-phase curriculum:

1. **Supervised pretraining.** Cross-entropy (action) + MSE (value) on exhaustively-searched optimal circuits.
2. **RL fine-tuning (PPO or SAC).** GAE [Sch+16] (γ=0.99, λ=0.95); clipped objective (ϵ=0.2); entropy bonus (c_{ent}=0.02). Advance complexity when success rate $> 60\%$.

Hyperparameters: LR $= 10^{-5}$; batch 4096; 10 PPO epochs/iter; 2000 total iterations. Trained on AWS EC2 + NVIDIA Tesla T4 (16 GB).

Results & Conclusion

Success rates on held-out polynomials ($n = 2$ variables):



- **SAC** leads on two-variable targets, learning effective symbolic construction strategies.
- **PPO+MCTS** scales better to three variables; explicit planning becomes increasingly important with dimensionality.
- **Curriculum** is essential for higher complexity; **sparse rewards** remain the primary bottleneck.

Future work: Scale to degree-8+ polynomials; stronger exploration strategies; AlphaTensor-style [Faw+22] integration.

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